



Lab Report 8 - Cybersecurity Analytics

# **Machine Learning in Cybersecurity data Analysis**

## Introduction

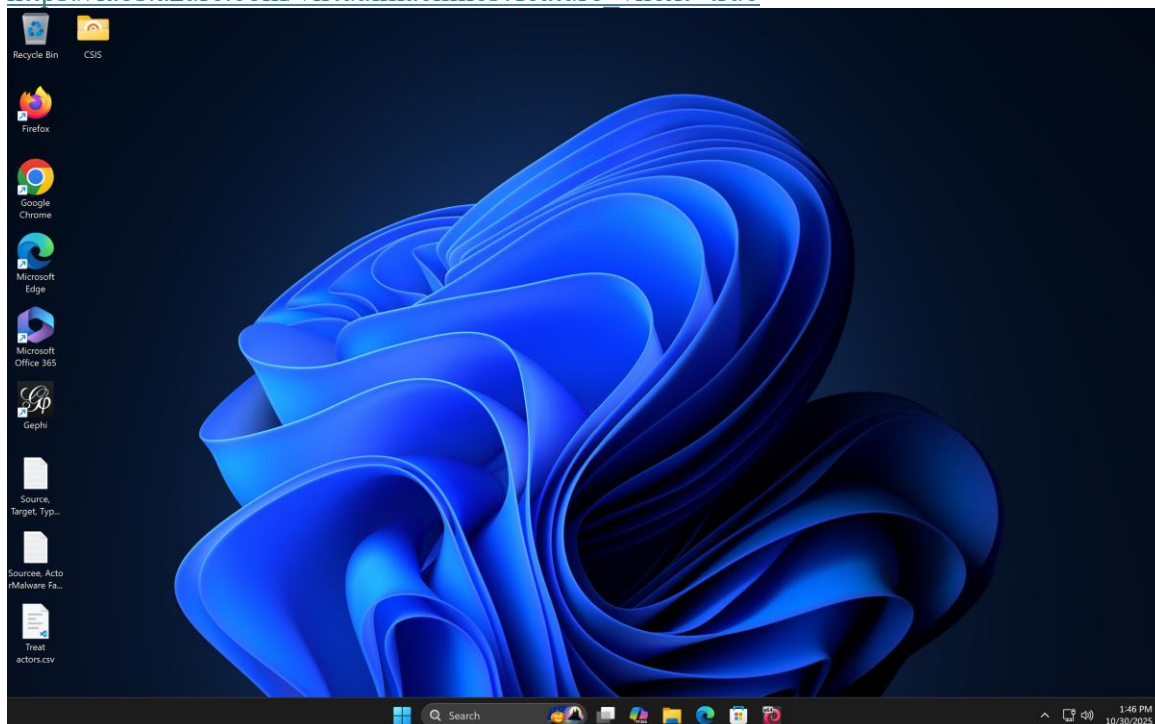
This lab is designed to work with CSIS-Admin VM machine to verify and set up a complete Python programming environment. First, CSIS-Admin virtual machine was configured for Python-based data analysis and machine learning workflows. Visual studio code opened from CSIS folder and created CSIS672-Hashan workspace folder to save all the upcoming development files. A Python virtual environment was configured within VS Code to isolate dependencies and prevent library conflicts. Python extension (Python extension by Windows) was installed and this extension also installed other extensions like Pylance and Python debugger. These extensions enable intelligent code completion, error highlighting, debugging capabilities, and execution directly within VS Code.

After setting up the environment, libraries like *Pandas* and *Matplotlib* were installed. *Pandas* enables powerful data manipulation and analysis, to perform some lightweight data wrangling (Chio & Freeman, 2018). The *Matplotlib* is widely used for creating visualizations. Machine Learning Data Handling was tested using the *payment\_fraud.csv* dataset, confirming that the environment could successfully import, read, and display structured data.

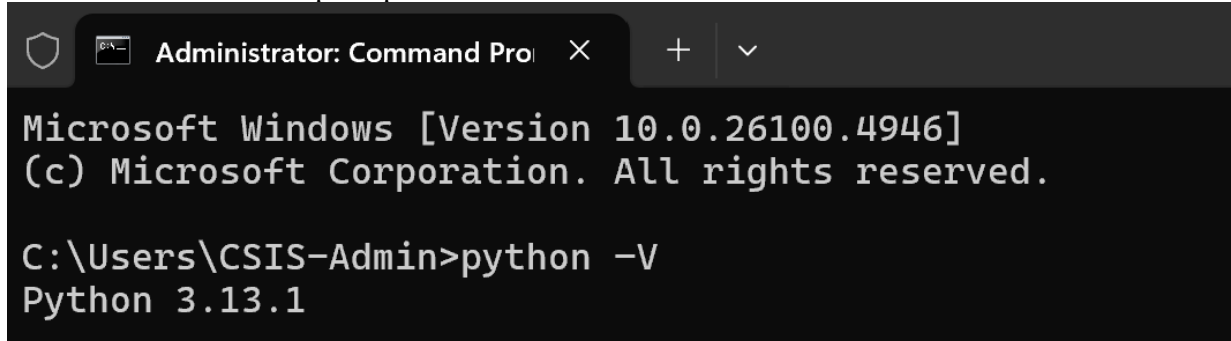
## Part 1 - Verify Python Installation

1. Logged to Azure Lab environment using following web link and logged into the Kali VM using Oracle VirtualBox.

[https://labs.azure.com/virtualmachines?feature\\_vnext=true](https://labs.azure.com/virtualmachines?feature_vnext=true)



2. Verified that the Python is installed in CSIS-Admin virtual machine by using `python -V` command on command prompt.

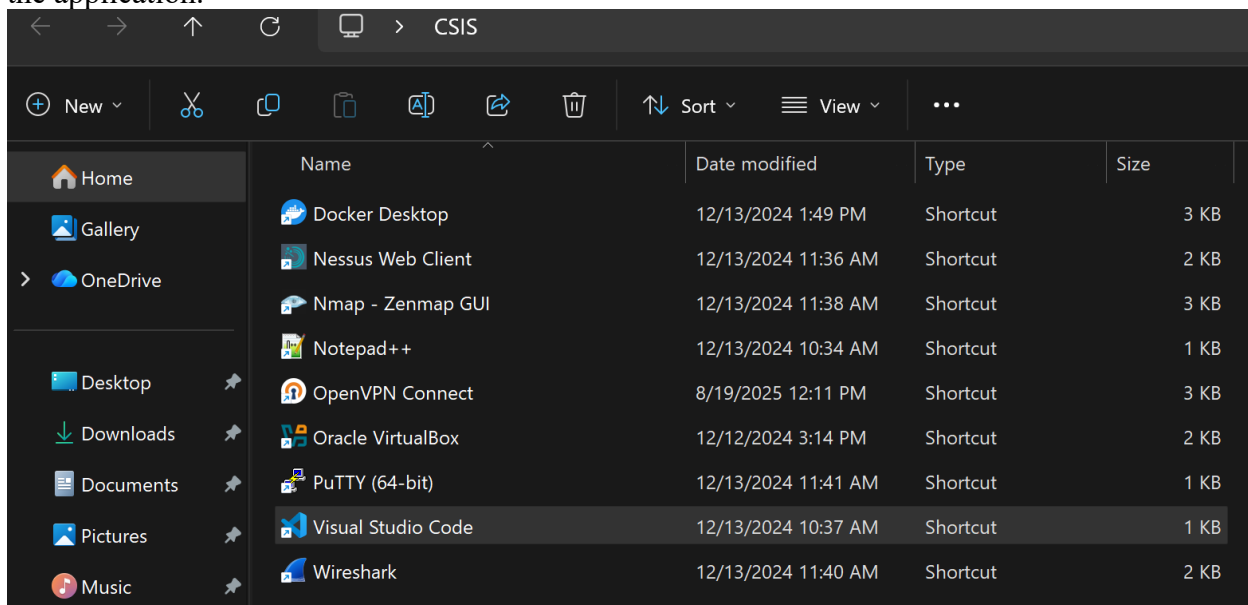


```
Microsoft Windows [Version 10.0.26100.4946]
(c) Microsoft Corporation. All rights reserved.

C:\Users\CSIS-Admin>python -V
Python 3.13.1
```

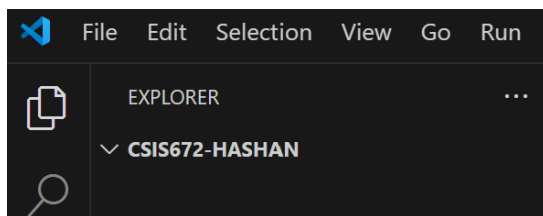
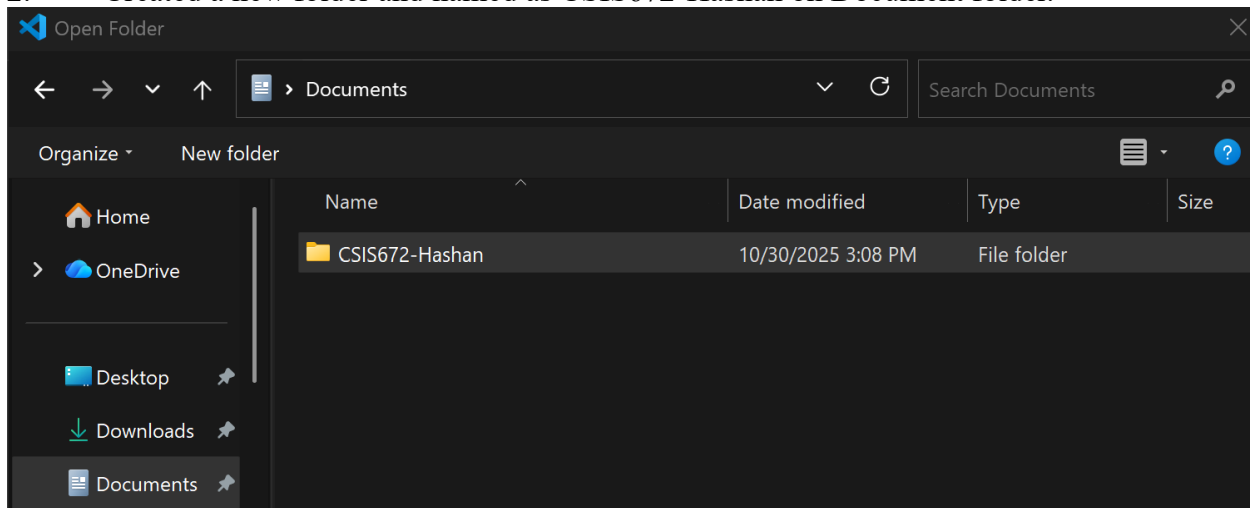
## Part 2 – VS Code

1. Located VS Code under the folder CSIS (C:\Users\CSIS-Admin\Desktop\CSIS) and run the application.

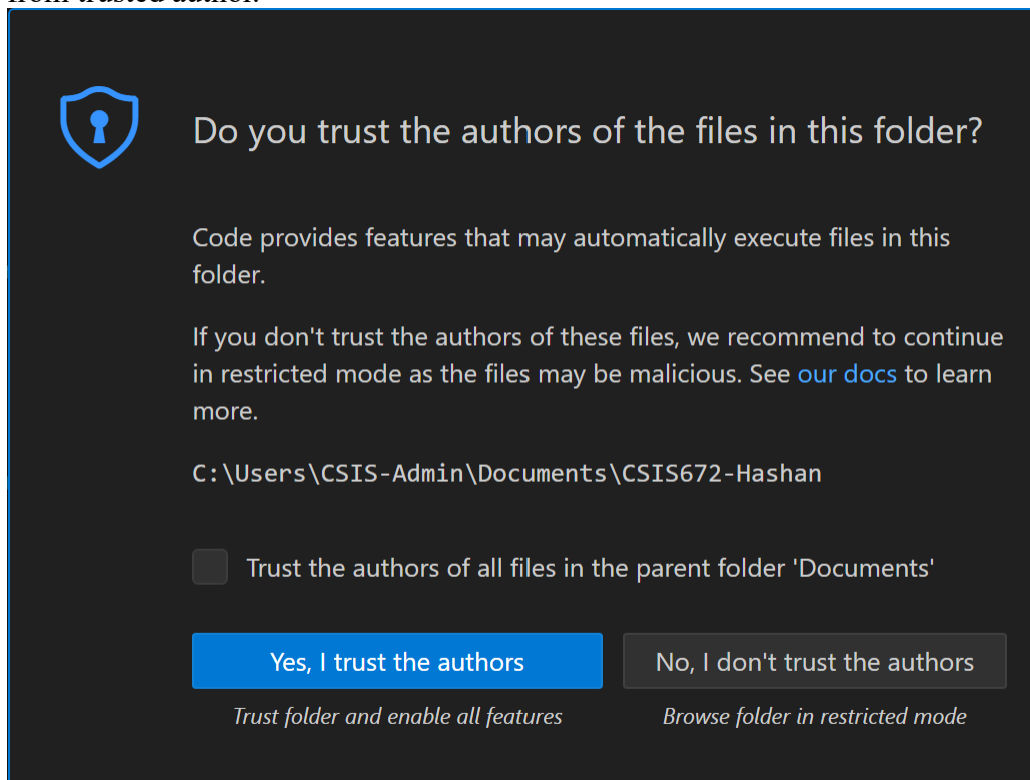


|           | Name               | Date modified       | Type     | Size |
|-----------|--------------------|---------------------|----------|------|
| Home      | Docker Desktop     | 12/13/2024 1:49 PM  | Shortcut | 3 KB |
| Gallery   | Nessus Web Client  | 12/13/2024 11:36 AM | Shortcut | 2 KB |
| OneDrive  | Nmap - Zenmap GUI  | 12/13/2024 11:38 AM | Shortcut | 3 KB |
|           | Notepad++          | 12/13/2024 10:34 AM | Shortcut | 1 KB |
| Desktop   | OpenVPN Connect    | 8/19/2025 12:11 PM  | Shortcut | 3 KB |
| Downloads | Oracle VirtualBox  | 12/12/2024 3:14 PM  | Shortcut | 2 KB |
| Documents | PuTTY (64-bit)     | 12/13/2024 11:41 AM | Shortcut | 1 KB |
| Pictures  | Visual Studio Code | 12/13/2024 10:37 AM | Shortcut | 1 KB |
| Music     | Wireshark          | 12/13/2024 11:40 AM | Shortcut | 2 KB |

2. Created a new folder and named as CSIS672-Hashan on Document folder.

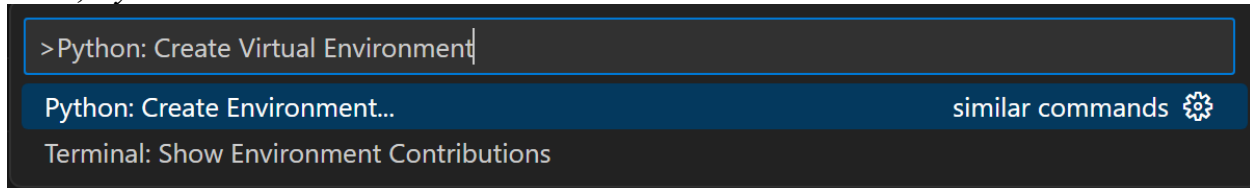


3. Added the folder to the workplace and clicked Yes to acknowledge that the added folder from trusted author.

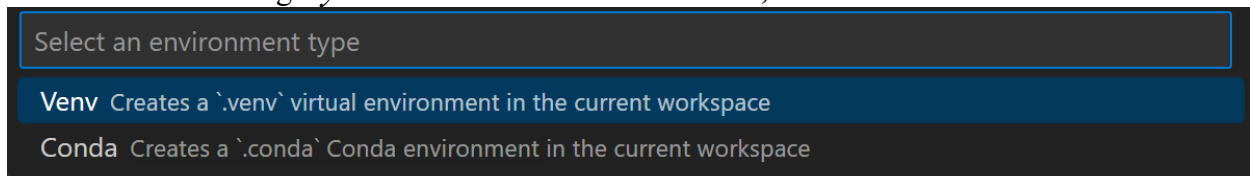


### **Part 3 – Creating a Virtual Environment in VS Code**

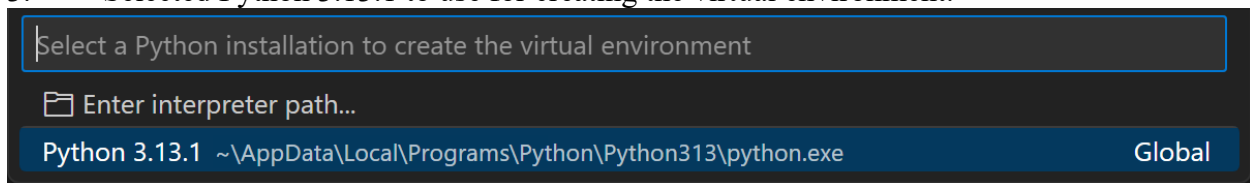
1. Opened the Command Palette by pressing Ctrl+Shift+P. Then enter the search term, *Python: Create Virtual Environment*



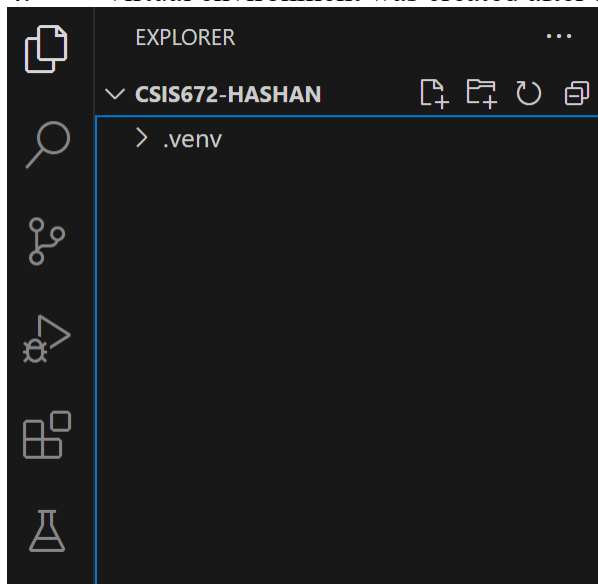
2. After selecting *Python: Create Environment...*, selected *Venv Creates .....*



3. Selected Python 3.13.1 to use for creating the virtual environment.

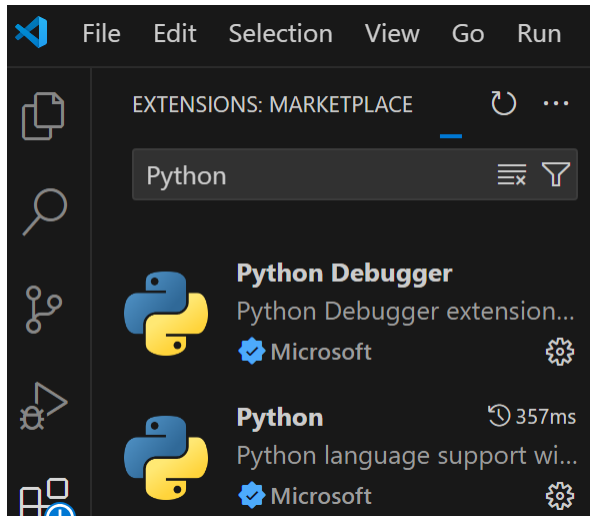


4. Virtual environment was created after a short time.



## Part 4 - Installing Extensions

1. Clicked on the extension icon and search Python extensions.

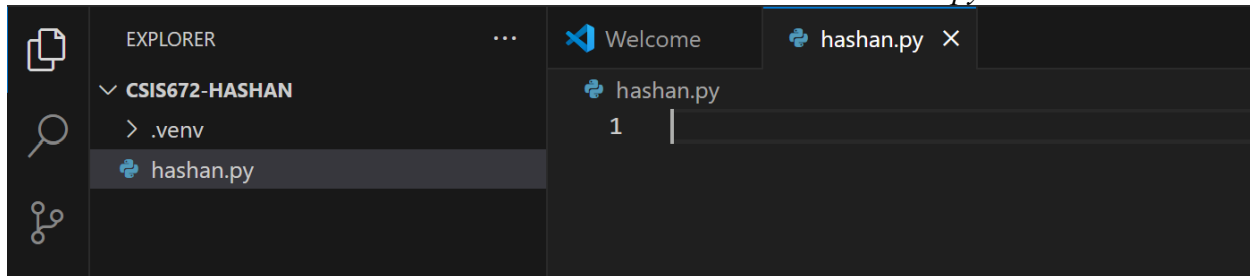


2. Installed the Python extension (Python extension by Windows). This extension also installed other extensions like pylance and python debugger.

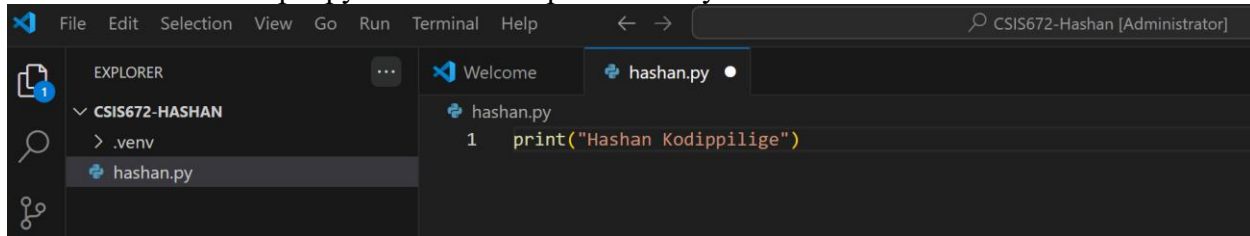


## Part 5 - Testing the New Environment

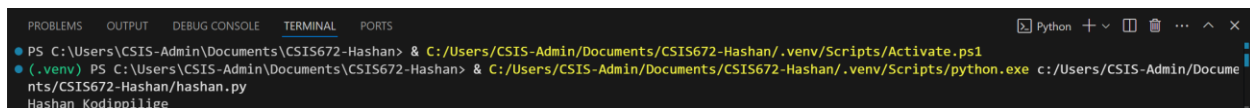
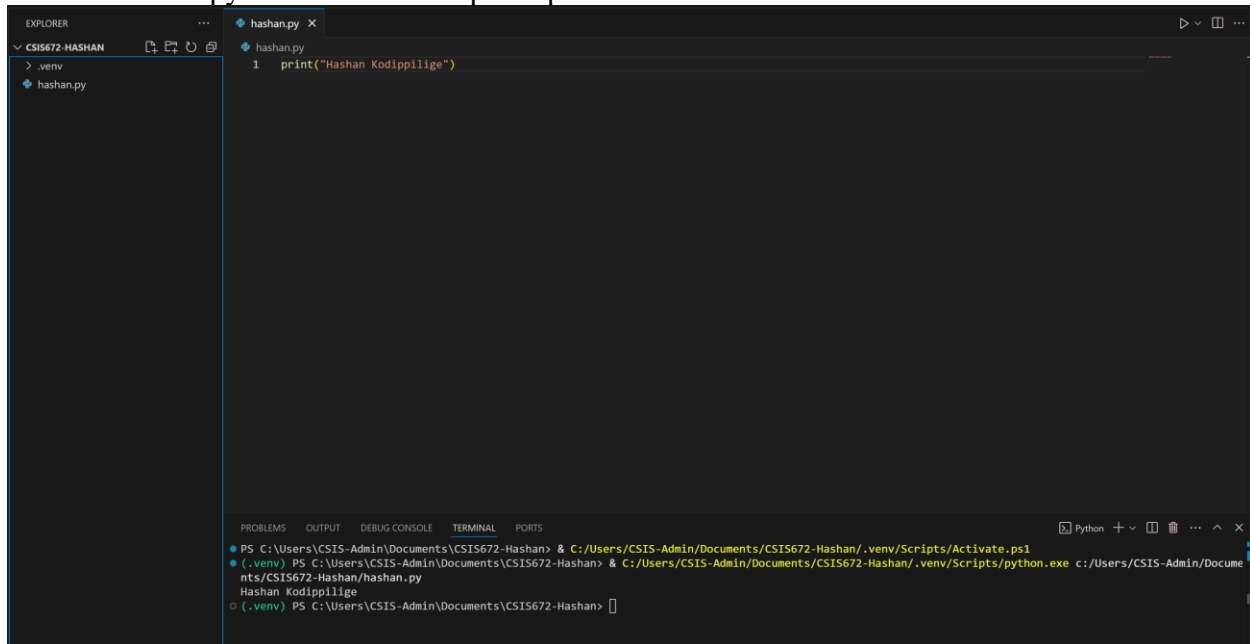
1. Created new file under CSIS672-HASHAN and named as *hashan.py*



2. Created a simple python code that prints out my first and last name.



3. Ran the python code and output captured in the Terminal window.



## Part 6 - Installing Libraries

1. Used *pip install panda* command to install the *pandas* library

```
(.venv) PS C:\Users\CSIS-Admin\Documents\CSIS672-Hashan> pip install pandas
```

2. Successfully installed the *Pandas* library.

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
● (.venv) PS C:\Users\CSIS-Admin\Documents\CSIS672-Hashan> pip install pandas
Collecting pandas
  Downloading pandas-2.3.3-cp313-cp313-win_amd64.whl.metadata (19 kB)
Collecting numpy>=1.26.0 (from pandas)
  Downloading numpy-2.3.4-cp313-cp313-win_amd64.whl.metadata (60 kB)
Collecting python-dateutil>=2.8.2 (from pandas)
  Downloading python_dateutil-2.9.0.post0-py2.py3-none-any.whl.metadata (8.4 kB)
Collecting pytz>=2020.1 (from pandas)
  Downloading pytz-2025.2-py2.py3-none-any.whl.metadata (22 kB)
Collecting tzdata>=2022.7 (from pandas)
  Downloading tzdata-2025.2-py2.py3-none-any.whl.metadata (1.4 kB)
Collecting six>=1.5 (from python-dateutil>=2.8.2->pandas)
  Downloading six-1.17.0-py2.py3-none-any.whl.metadata (1.7 kB)
Downloading pandas-2.3.3-cp313-cp313-win_amd64.whl (11.0 MB)
  ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 11.0/11.0 MB 38.4 MB/s 0:00:00
Downloading numpy-2.3.4-cp313-cp313-win_amd64.whl (12.8 MB)
  ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 12.8/12.8 MB 59.1 MB/s 0:00:00
Downloading python_dateutil-2.9.0.post0-py2.py3-none-any.whl (229 kB)
Downloading pytz-2025.2-py2.py3-none-any.whl (509 kB)
Downloading six-1.17.0-py2.py3-none-any.whl (11 kB)
Downloading tzdata-2025.2-py2.py3-none-any.whl (347 kB)
Installing collected packages: pytz, tzdata, six, numpy, python-dateutil, pandas
Successfully installed numpy-2.3.4 pandas-2.3.3 python-dateutil-2.9.0.post0 pytz-2025.2 six-1.17.0 tzdata-2025.2
○ (.venv) PS C:\Users\CSIS-Admin\Documents\CSIS672-Hashan>

```

3. Installed the *Matplotlib* library.

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Python + - [ ] ... [ ] [ ]

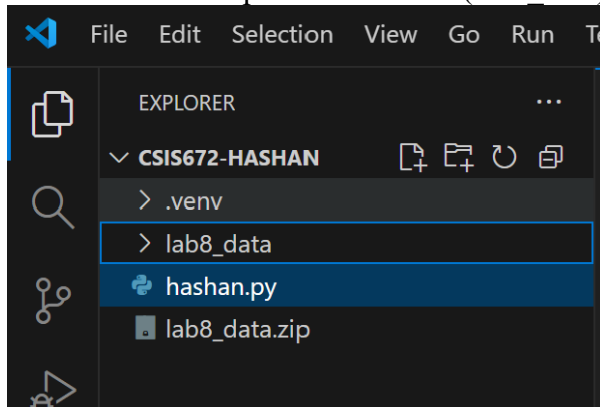
PS C:\Users\CSIS-Admin\Documents\CSIS672-Hashan> pip install Matplotlib
  Downloading kiwisolver-1.4.9-cp313-cp313-win_amd64.whl.metadata (6.4 kB)
Collecting numpy>=1.23 (from Matplotlib)
  Using cached numpy-2.3.4-cp313-cp313-win_amd64.whl.metadata (60 kB)
Collecting packaging>=20.0 (from Matplotlib)
  Downloading packaging-25.0-py3-none-any.whl.metadata (3.3 kB)
Collecting pillow>=8 (from Matplotlib)
  Downloading pillow-12.0.0-cp313-cp313-win_amd64.whl.metadata (9.0 kB)
Collecting pyparsing>=3 (from Matplotlib)
  Downloading pyparsing-3.2.5-py3-none-any.whl.metadata (5.0 kB)
Collecting python-dateutil>=2.7 (from Matplotlib)
  Using cached python_dateutil-2.9.0.post0-py2.py3-none-any.whl.metadata (8.4 kB)
Collecting six>=1.5 (from python-dateutil>=2.7->Matplotlib)
  Using cached six-1.17.0-py2.py3-none-any.whl.metadata (1.7 kB)
Downloading matplotlib-3.10.7-cp313-cp313-win_amd64.whl (8.1 MB)
  8.1/8.1 MB 32.9 MB/s eta 0:00:00
Downloading contourpy-1.3.3-cp313-cp313-win_amd64.whl (226 kB)
Downloading cycler-0.12.1-py3-none-any.whl (8.3 kB)
Downloading fonttools-4.60.1-cp313-cp313-win_amd64.whl (2.3 MB)
  2.3/2.3 MB 35.7 MB/s eta 0:00:00
Downloading kiwisolver-1.4.9-cp313-cp313-win_amd64.whl (73 kB)
Using cached numpy-2.3.4-cp313-cp313-win_amd64.whl (12.8 MB)
Downloading packaging-25.0-py3-none-any.whl (66 kB)
Downloading pillow-12.0.0-cp313-cp313-win_amd64.whl (7.0 MB)
  7.0/7.0 MB 71.5 MB/s eta 0:00:00
Downloading pyparsing-3.2.5-py3-none-any.whl (113 kB)
Using cached python_dateutil-2.9.0.post0-py2.py3-none-any.whl (229 kB)
Using cached six-1.17.0-py2.py3-none-any.whl (11 kB)
Installing collected packages: six, pyparsing, pillow, packaging, numpy, kiwisolver, fonttools, cycler, python-dateutil, contourpy, Matplotlib
Successfully installed Matplotlib-3.10.7 contourpy-1.3.3 cycler-0.12.1 fonttools-4.60.1 kiwisolver-1.4.9 numpy-2.3.4 packaging-25.0 pillow-12.0.0 pyparsing-3.2.5 python-dateutil-2.9.0.post0 six-1.17.0

[notice] A new release of pip is available: 24.3.1 -> 25.3
[notice] To update, run: python.exe -m pip install --upgrade pip
PS C:\Users\CSIS-Admin\Documents\CSIS672-Hashan>

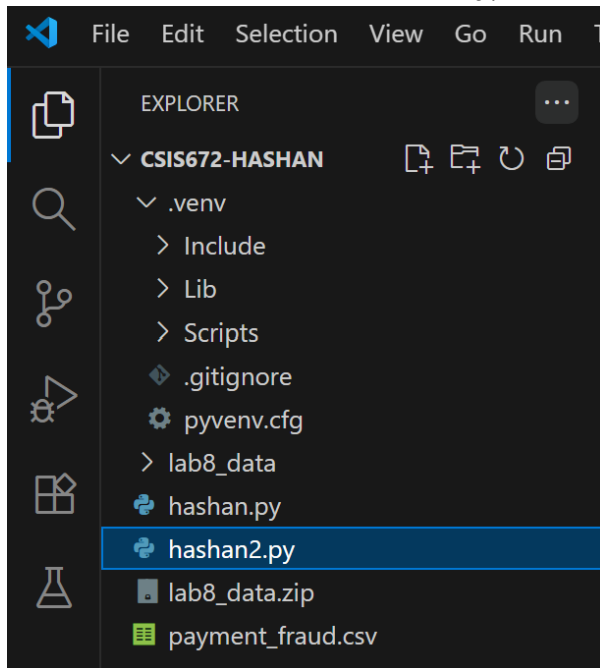
```

## **Part 7 - Using Machine Learning Libraries**

1. Placed the provided dataset (lab8\_data) in the CSIS672-HASHAN folder.



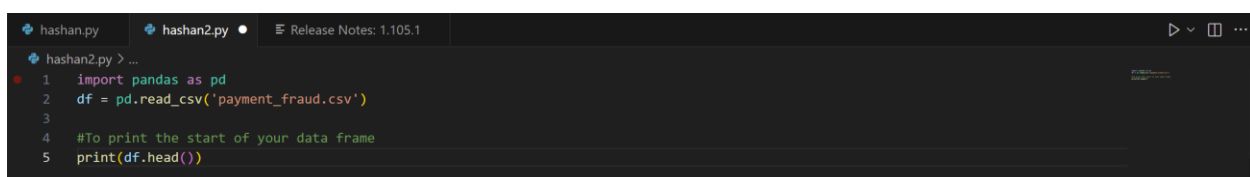
2. Created new file under CSIS672-HASHAN and named as hashan2.py



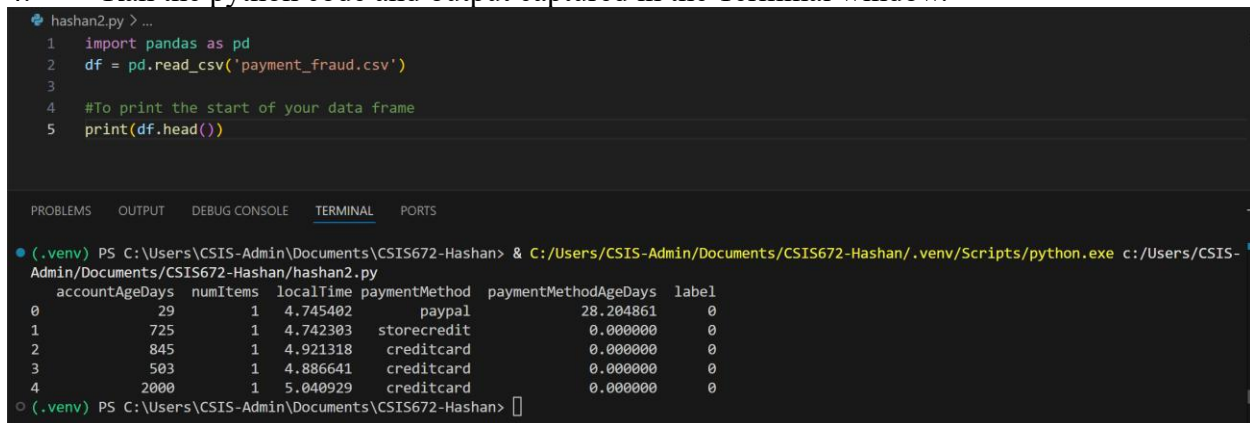
3. Placed the flowing code in the file.

```
import pandas as pd
df = pd.read_csv('payment_fraud.csv')

#To print the start of your data frame
print(df.head())
```



4. Ran the python code and output captured in the Terminal window.



```
hashan2.py > ...
1 import pandas as pd
2 df = pd.read_csv('payment_fraud.csv')
3
4 #To print the start of your data frame
5 print(df.head())
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
(.venv) PS C:\Users\CSIS-Admin\Documents\CSIS672-Hashan> & C:/Users/CSIS-Admin/Documents/CSIS672-Hashan/.venv/Scripts/python.exe c:/Users/CSIS-Admin/Documents/CSIS672-Hashan/hashan2.py
accountAgeDays numItems localTime paymentMethod paymentMethodAgeDays label
0 29 1 4.745402 paypal 28.204861 0
1 725 1 4.742303 storecredit 0.000000 0
2 845 1 4.921318 creditcard 0.000000 0
3 503 1 4.886641 creditcard 0.000000 0
4 2000 1 5.040929 creditcard 0.000000 0
(.venv) PS C:\Users\CSIS-Admin\Documents\CSIS672-Hashan>
```

The terminal displays the first five rows of the dataset. The displayed data shows the account ages, local time, and a mix of payment methods, payment method age days. All the transactions currently have label 0.

## Conclusion

This lab successfully established a fully functional Python development environment designed for data analysis and machine learning applications. Verifying the existing Python installation (Python 3.13.1) using the `python -V` command establishes the foundational step for all next configurations. Creating *CSIS672-Hashan* folder will isolate lab files, datasets, and scripts within a structured workspace. Installing key libraries such as Pandas and Matplotlib further prepared the environment for handling real-world datasets, enabling efficient data manipulation and visualization. Pandas supports efficient handling of structured data and advanced data wrangling, while Matplotlib enables high-quality data visualization for exploratory analysis.

The dataset named *payment\_fraud.csv* was loaded using Pandas displays five rows of data set with relevant details related to each data set. We can improve the analyzing data sets by improving our Python codes and use of more libraries.

## References

1. Chio, C., & Freeman, D. (2018). *Machine Learning and Security: Protecting Systems with Data and Algorithms*. O'Reilly Media.
2. GeeksforGeeks. (2025, July 11). *Pandas Read CSV in Python*. Sanchhaya Education Private Limited. [https://www.geeksforgeeks.org/pandas/python-read-csv-using-pandas-read\\_csv/](https://www.geeksforgeeks.org/pandas/python-read-csv-using-pandas-read_csv/)
3. Microsoft Docs. (2025). *Python in Visual Studio Code*. Microsoft. Retrieved October 28, 2025, from <https://code.visualstudio.com/docs/languages/python>
4. McKinney, W. (2022). *Python for Data Analysis: Data Wrangling with Pandas, NumPy, and Jupyter* (3rd ed.). O'Reilly Media
5. Python Software Foundation. (2024). *Python 3.13.1 Documentation*. Retrieved from October 28, 2025, from <https://www.python.org/downloads/release/python-3131/>